

Wuao Liu

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Education

University of Massachusetts Amherst

Ph.D. in Computer Science

Amherst, MA

Aug 2024 – Dec 2028 (expected)

University of Michigan, Ann Arbor

M.S. in Robotics

Ann Arbor, MI

Aug 2021 – Apr 2023

Zhejiang University (ZJU)

B.Eng. in Robotics

Hangzhou, China

Sep 2017 – Jun 2021

Work Experience

LinkedIn

AI/ML Engineer Intern (Generative AI)

Mountain View, CA

May 2026 – Aug 2026

- Implemented an LLM-based pipeline to extract job-relevant attributes from member profiles for job matching.

Honda Research Institute

Research Scientist Intern (Multimodal ML)

San Jose, CA

May 2025 – Aug 2025

- Built a multimodal framework integrating spatial audio cues with LLMs for AR/VR scene understanding.
- Implemented spatial audio data generation and filtering pipelines for a large-scale dataset with **872k** perception and reasoning question-answer pairs.

University of Michigan, Ann Arbor

Research Engineer

Ann Arbor, MI

Jun 2023 – Apr 2024

- Designed an egocentric video collection protocol and curated a dataset of **224** instructional cooking videos.
- Trained a Faster R-CNN object detector for **14** kitchen items, achieving **0.86** mAP on the curated video dataset.
- Developed a procedural step classification model achieving **92.54%** accuracy, enabling automated detection of step-transition errors.

Tencent AI Lab

Machine Learning Engineer Intern

Shenzhen, China

Jun 2021 – Jul 2021

- Implemented a deep Q-learning-based electric vehicle charging navigation method which jointly considered traffic conditions, charging prices, and wait times.

Publications

- [1] **Wuao Liu**, M. Chasmai, S. Maji, G. Van Horn. *Masked Autoencoders with Limited Data: Does It Work? A Fine-Grained Bioacoustics Case Study*. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshop (FGVC13), 2026. **Oral Presentation** [\[paper\]](#)
- [2] L. Lawrence, O. Saha, R. Daroya, M. Chasmai, **Wuao Liu**, et al. *RealBirdID: Benchmarking Bird Species Identification in the Era of MLLMs*. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026. [\[paper\]](#)
- [3] M. Chasmai, **Wuao Liu**, S. Maji, G. Van Horn. *Bioacoustic Geolocation: Species Sounds as Geographic Signals*. International Conference on Machine Learning (ICML), 2026. [\[paper\]](#)
- [4] J. Zhong, Y. Lai, M. Li, Y. Li, **Wuao Liu**, et al. *From Shortcuts to Reasoning: Robust Post-Training of Theory of Mind with Reinforcement Learning*. International Conference on Machine Learning (ICML), 2026. [\[paper\]](#)
- [5] F. Bellos, Y. Li, **Wuao Liu**, J. J. Corso. *Can Large Language Models Reason About Goal-Oriented Tasks?* Annual Meeting of the Association for Computational Linguistics (ACL) Workshop (SCALE-LLM), 2024. [\[paper\]](#)

Technical Skills

Programming: Python; C++; MATLAB; $\text{\LaTeX}$; ROS

Deep Learning: PyTorch; TensorFlow 2; HuggingFace Ecosystem